Practical Demo: Constructing a Benchmark-Aware Bond Portfolio

From cash to a cell-matched portfolio and a multi-factor portfolio

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The investment decision

A manager receives **$50 million in cash and owns no bonds**. Construct a benchmark-aware bond portfolio in two ways:

1. **Cell-based construction:** match the benchmark's sector/maturity buckets, then select bonds within each bucket at the lowest acquisition cost.
2. **Multi-factor construction:** choose bond weights that minimize predicted tracking error using Treasury key-rate and spread-factor exposures.

Both methods use the same benchmark, investable universe, cash budget, and portfolio limits. The cell method additionally requires exact cell weights. There are no starting bond holdings, sales, or rebalancing turnover penalties. Every nonzero position is an initial purchase.

All bond descriptions, analytics, spreads, costs, and risk estimates below are **illustrative inputs**, not market observations or executable quotes. Weights refer to the bond sleeve after purchase costs; all amounts are in US dollars.

1 Specify the mandate

Requirement	Limit
Initial cash, including acquisition costs	$50 million
Portfolio positions	Long-only; bond-sleeve weights sum to 100%
Maximum weight in one bond	18% of the bond sleeve
Minimum Treasury weight	25%
Maximum corporate weight	50%
Duration versus benchmark	Within 0.05 years
Weighted yield proxy versus benchmark	At least 2 bp higher
Acquisition cost	At most 6 bp of invested bond market value

The modest yield-proxy floor gives both constructors a deliberate active objective. It is not a forecast of excess return. No hard tracking-error limit is imposed: both portfolios are evaluated with the same model, and the multi-factor constructor minimizes that measure.

1.1 Account for purchase costs inside the cash budget

Let B be initial cash, w_i bond-sleeve weights, and c_i each bond's purchase cost per dollar of market value. The amount invested in bonds is

$$Q = \frac{B}{1 + \sum_i w_i c_i}.$$

Buy Qw_i of each bond and pay $Qw_i c_i$ of acquisition costs. Therefore

$$\sum_i Qw_i + \sum_i Qw_i c_i = B.$$

No additional cash is assumed. Costs reduce the initial amount invested; they are a one-time wealth drag, not annual tracking-error volatility.

2 Define the investable bonds and benchmark

The benchmark is an explicitly supplied set of bond weights. It is a reference index, not an existing client portfolio. Both constructors may buy any of the 12 securities below. Each corporate label represents a different issuer, so bond concentration and issuer concentration coincide within this universe.

All bonds are treated as dollar-denominated, option-free securities. Yields are annual yield proxies. Spread means the supplied spread above the reference Treasury curve, measured consistently across bonds; Treasuries have zero spread. The market-value-weighted spread is an average-spread proxy, not a separately solved portfolio option-adjusted spread.

2.1 Benchmark weights and classification

Bond	Sector	Maturity (years)	Benchmark (%)
UST 2Y	Treasury	2	10.00
UST 5Y	Treasury	5	13.00
UST 10Y	Treasury	10	10.00
Agency 3Y	Agency	3	9.00
Agency 7Y	Agency	7	10.00
Bank A 3Y	Corporate	3	7.00
Bank B 5Y	Corporate	5	9.00
Utility A 7Y	Corporate	7	8.00
Industrial A 10Y	Corporate	10	7.00
Industrial B 5Y	Corporate	5	7.00
Telecom A 10Y	Corporate	10	5.00
Consumer A 7Y	Corporate	7	5.00

2.2 Yields, spreads, and acquisition costs

Purchase costs are paid once on the acquired market value, beyond that market value. They include the assumed execution friction; no additional spread or fee is added elsewhere in the budget.

Bond	Yield (%)	Spread (bp)	Trading cost (bp)
UST 2Y	4.15	0	1
UST 5Y	4.35	0	2
UST 10Y	4.55	0	3
Agency 3Y	4.42	38	3
Agency 7Y	4.70	55	5
Bank A 3Y	5.05	92	6
Bank B 5Y	5.28	112	7
Utility A 7Y	5.55	138	9
Industrial A 10Y	5.85	165	12
Industrial B 5Y	5.40	125	8
Telecom A 10Y	6.00	182	14
Consumer A 7Y	5.62	148	10

2.3 Treasury and spread sensitivities

Key-rate durations and spread durations are measured in years. Convexity is measured in years squared. For each bond, the three key-rate durations sum to its effective duration in this simplified model.

Bond	KRD 2	KRD 5	KRD 10	Spread duration
UST 2Y	1.70	0.18	0.00	0.00
UST 5Y	0.35	3.75	0.35	0.00
UST 10Y	0.05	1.20	6.60	0.00
Agency 3Y	2.30	0.42	0.00	2.60
Agency 7Y	0.25	4.30	1.27	5.65
Bank A 3Y	2.20	0.45	0.00	2.55
Bank B 5Y	0.45	3.35	0.55	4.20
Utility A 7Y	0.20	3.80	1.65	5.45
Industrial A 10Y	0.10	1.25	6.00	7.05
Industrial B 5Y	0.50	3.10	0.58	4.02
Telecom A 10Y	0.10	1.10	5.90	6.75
Consumer A 7Y	0.25	3.65	1.55	5.25

Bond	Duration	Convexity
UST 2Y	1.88	4.80
UST 5Y	4.45	23.50
UST 10Y	7.85	72.00
Agency 3Y	2.72	9.00
Agency 7Y	5.82	39.50
Bank A 3Y	2.65	8.50
Bank B 5Y	4.35	22.00
Utility A 7Y	5.65	37.00
Industrial A 10Y	7.35	62.00
Industrial B 5Y	4.18	20.50
Telecom A 10Y	7.10	58.00
Consumer A 7Y	5.45	35.00

3 Identify the four risk factors

A **factor is a market change**. A duration is a security's **sensitivity** to that change. The risk model contains exactly four common factors:

Factor	Change being modeled	Bond sensitivity
2Y Treasury yield	Change at the 2-year Treasury node	2Y key-rate duration
5Y Treasury yield	Change at the 5-year Treasury node	5Y key-rate duration
10Y Treasury yield	Change at the 10-year Treasury node	10Y key-rate duration
Common spread	Change in the common corporate-spread driver	Spread duration times sector beta

The spread-driver beta is **0 for Treasuries, 0.25 for agencies, and 1 for corporates**. A +100 bp common-spread shock therefore means unchanged Treasury spreads, +25 bp agency spreads, and +100 bp corporate spreads. This same scaling is used in both predicted risk and scenarios.

There is no separate level factor on top of the three key rates. A parallel Treasury move means applying the same change to all three nodes; adding another level shock would count that move twice.

3.1 Return mapping and units

For changes measured in basis points, the first-order bond return in basis points is

$$r_i \approx -KRD_{i,2}\Delta y_2 - KRD_{i,5}\Delta y_5 - KRD_{i,10}\Delta y_{10} - \beta_i SD_i \Delta s + \epsilon_i.$$

For example, a bond with 5Y key-rate duration 3.75 loses approximately **37.5 bp** when only the 5Y Treasury yield rises 10 bp. A corporate bond with spread duration 4.20 loses approximately **42 bp** when the common spread rises 10 bp, with Treasury yields unchanged.

Here ϵ_i is security-specific return. Residuals are assumed mutually independent and independent of all four common factors. The model omits additional rating, sector, liquidity, currency, and option factors.

3.2 Factor volatilities, correlations, and residual risk

All volatilities are **annualized**, and predicted tracking error is annualized. Factor volatilities below are volatilities of yield/spread changes in bp per square-root year; residual volatilities are bond-return volatilities in the same time units. They are supplied inputs, not estimates from the bond table.

Factor	Annual shock volatility (bp)
2Y Treasury yield	70.0
5Y Treasury yield	80.0
10Y Treasury yield	90.0
Common spread	60.0

Factor	2Y	5Y	10Y	Spread
2Y	1.00	0.82	0.62	0.05
5Y	0.82	1.00	0.85	0.08
10Y	0.62	0.85	1.00	0.10
Spread	0.05	0.08	0.10	1.00

Residual group	Annual residual return volatility (bp)
Treasury / agency	30
Corporate	100

The factor covariance matrix is $\Sigma_F = \mathrm{diag}(\sigma_F)\rho\,\mathrm{diag}(\sigma_F)$, in squared basis points. Correlations are assumed fixed for this exercise. In practice they require a separate historical or forward-looking risk-estimation process; security terms and benchmark weights alone do not determine them.

4 Map holdings into active risk

Let L contain the negative duration loadings in the return equation and let $a = w - w_B$ be active bond-sleeve weights. Predicted active-return variance is

$$TE^2 = (a'L)\Sigma_F(L'a) + \sum_i a_i^2 \sigma_{\epsilon,i}^2.$$

The square root is annual TE in basis points. For example, increasing one bond's weight by 10 percentage points adds one tenth of its factor sensitivities and residual exposure, before accounting for the offsetting funding position.

This comparison measures the two normalized bond sleeves against the same normalized benchmark. Acquisition costs are reported separately as a known initial drag. The benchmark index itself has no acquisition cost.

5 Construct the two portfolios

5.1 Method 1: exact cell matching

Define cells by **sector** and **remaining maturity**. Sectors are Treasury, agency, and corporate; maturity buckets are $(0, 4]$, $(4, 8]$, and over 8 years. There are nine possible cells and eight occupied cells in this universe.

Cell	Benchmark %
Agency / 0-4Y	9.00
Agency / 4-8Y	10.00
Corporate / 0-4Y	7.00
Corporate / 4-8Y	29.00
Corporate / 8Y+	12.00
Treasury / 0-4Y	10.00
Treasury / 4-8Y	13.00
Treasury / 8Y+	10.00

For each occupied cell g, require

$$\sum_{i \in g} w_i = \sum_{i \in g} w_{B,i}.$$

Within those cell budgets, minimize $\sum_i c_i w_i$ subject to all common mandate limits. Duration and yield are guardrails; **the selection step does not use factor covariance or predicted tracking error**. A linear program makes the cheapest feasible selection within the exact cells. Its output is assessed with the shared risk model afterward.

5.2 Method 2: multi-factor risk minimization

Remove the exact-cell constraints and instead minimize the common model's predicted tracking-error variance:

$$\min_w \; TE(w)^2.$$

Keep every common mandate limit, including duration, concentration, sector, yield-proxy floor, and acquisition-cost cap. The optimizer can shift between cells when that improves the combined factor and residual risk. The benchmark itself has zero TE but fails the required 2 bp yield-proxy increment.

Both problems use long-only weights summing to one. Dividing risk variance by 10,000 in the calculation is only numerical scaling; it does not change the solution. The initial feasible point is a cheapest feasible purchase plan, not a set of pre-existing holdings.

6 Compare the constructed portfolios

6.1 Bond allocations

All portfolio weights below are percentages of their invested bond sleeve. A zero weight means the constructor does not buy that security.

Bond	Benchmark (%)	Cell (%)	Multi-factor (%)
UST 2Y	10.00	10.00	6.67
UST 5Y	13.00	13.00	13.01
UST 10Y	10.00	10.00	10.65
Agency 3Y	9.00	9.00	11.01
Agency 7Y	10.00	10.00	8.66
Bank A 3Y	7.00	7.00	8.25
Bank B 5Y	9.00	0.00	9.02
Utility A 7Y	8.00	17.45	7.62
Industrial A 10Y	7.00	2.98	5.92
Industrial B 5Y	7.00	11.55	8.19
Telecom A 10Y	5.00	9.02	5.51
Consumer A 7Y	5.00	0.00	5.48

6.2 Verify cell matching

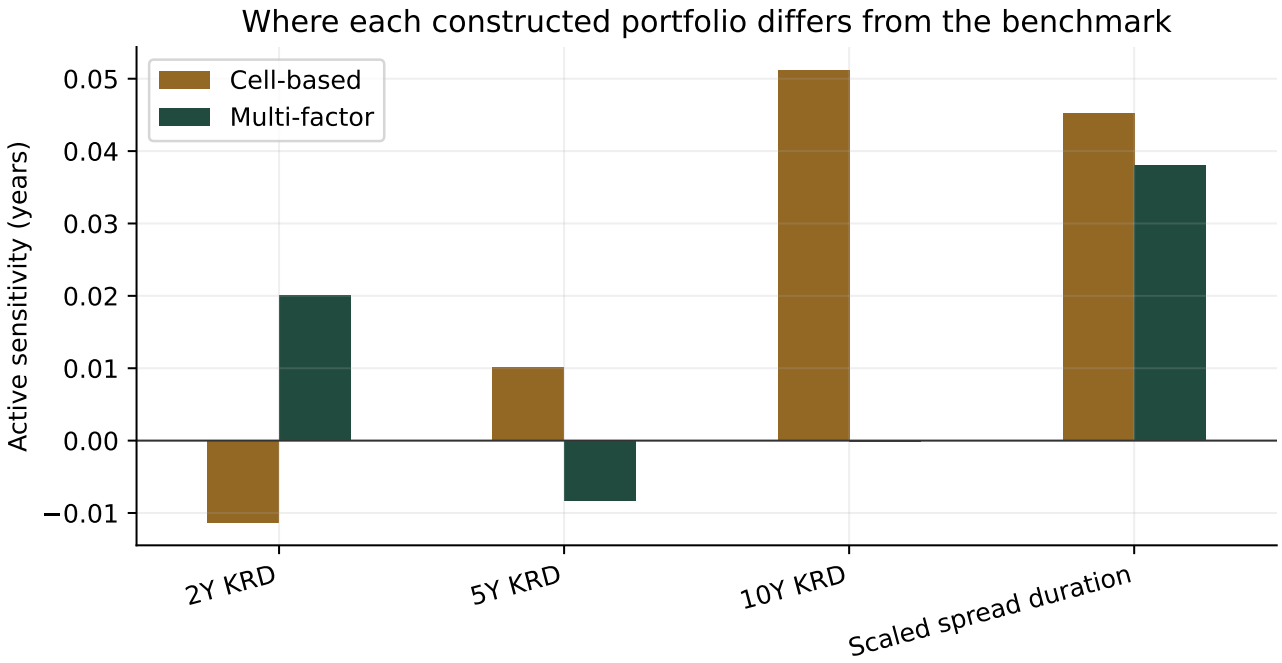
Cell	Benchmark %	Cell %	Factor %
Agency / 0-4Y	9.00	9.00	11.01
Agency / 4-8Y	10.00	10.00	8.66
Corporate / 0-4Y	7.00	7.00	8.25
Corporate / 4-8Y	29.00	29.00	30.31
Corporate / 8Y+	12.00	12.00	11.44
Treasury / 0-4Y	10.00	10.00	6.67
Treasury / 4-8Y	13.00	13.00	13.01
Treasury / 8Y+	10.00	10.00	10.65

6.3 Compare exposure and risk

Portfolio	Yield proxy (%)	Duration (years)	Annual TE (bp)
Benchmark	4.944	4.842	0.000
Cell-based	4.965	4.892	16.782
Multi-factor	4.965	4.853	3.534

Portfolio	Treasury (%)	Corporate (%)	Largest bond (%)
Benchmark	33.00	48.00	13.00
Cell-based	33.00	48.00	17.45
Multi-factor	30.33	50.00	13.01

Portfolio	Average spread (bp)
Benchmark	73.28
Cell-based	75.21
Multi-factor	75.31



Exact cell matching does not imply exact factor matching. Bonds in one cell can have different key-rate durations, spread sensitivities, and residual risk. The multi-factor result has the smallest modeled TE among portfolios satisfying the common mandate. This does not establish higher future realized returns or robustness to a misspecified covariance model.

6.4 Explain the sources of predicted risk

The following amounts add in **variance**, not standard deviations. The factor contribution for factor k is $z_k(\Sigma_F z)_k$, where $z = a'L$. A contribution can be negative when covariance offsets other active risks.

Portfolio	Source	Variance (bp²)
Cell-based	2Y Treasury yield	-2.273
Cell-based	5Y Treasury yield	3.466
Cell-based	10Y Treasury yield	23.423
Cell-based	Common spread	8.713
Cell-based	Independent residuals	248.299
Multi-factor	2Y Treasury yield	1.342
Multi-factor	5Y Treasury yield	-0.434
Multi-factor	10Y Treasury yield	-0.009
Multi-factor	Common spread	5.239
Multi-factor	Independent residuals	6.352

7 Turn initial cash into purchase orders

These are **purchases from cash**, not differences from an existing bond portfolio. Bond market values plus acquisition costs must equal $50 million.

Market value is the full settlement value, including accrued interest. Converting it to face amount requires the full price per $100 face, feasible lots, and available inventory. No prices or lot-size assumptions are supplied here, so the table is a market-value allocation plan rather than executable face-amount tickets.

Cell-based initial purchase plan

Bond	Weight (%)	Buy $(m)\|Cost()$	
UST 2Y	10.000	4.997	499.702
UST 5Y	13.000	6.496	1299.225
UST 10Y	10.000	4.997	1499.106
Agency 3Y	9.000	4.497	1349.195
Agency 7Y	10.000	4.997	2498.510
Bank A 3Y	7.000	3.498	2098.748
Bank B 5Y	0.000	0.000	0.000
Utility A 7Y	17.446	8.718	7845.901
Industrial A 10Y	2.979	1.489	1786.376
Industrial B 5Y	11.554	5.774	4618.950
Telecom A 10Y	9.021	4.508	6310.887
Consumer A 7Y	0.000	0.000	0.000

Multi-factor initial purchase plan

Bond	Weight (%)	Buy $(m)\|Cost()$	
UST 2Y	6.670	3.333	333.303
UST 5Y	13.013	6.503	1300.515
UST 10Y	10.650	5.322	1596.581
Agency 3Y	11.007	5.500	1650.039
Agency 7Y	8.660	4.328	2163.784
Bank A 3Y	8.253	4.124	2474.409
Bank B 5Y	9.015	4.505	3153.455
Utility A 7Y	7.623	3.809	3428.441
Industrial A 10Y	5.921	2.959	3550.378
Industrial B 5Y	8.191	4.093	3274.440
Telecom A 10Y	5.515	2.756	3858.000
Consumer A 7Y	5.482	2.739	2739.497

Method	Bonds ()$\|Costs()$	Cash used ($)	
Cell-based	49,970,193.40	29,806.60	50,000,000.00
Multi-factor	49,970,477.16	29,522.84	50,000,000.00

8 Stress the completed portfolios

A parallel Treasury shock uses the three key rates together. For that special case, add the supplied parallel convexity correction:

$$\Delta P_i/P_i \approx -D_i\Delta y - \beta_i SD_i\Delta s + \tfrac{1}{2}C_i(\Delta y)^2.$$

For nonparallel curve twists, use first-order key-rate exposures. The inputs do not supply cross-key-rate convexities. The scenarios are instantaneous price changes, excluding carry, defaults, and reinvestment. No probabilities are assigned to them.

8.1 Gross price changes and wealth changes after costs

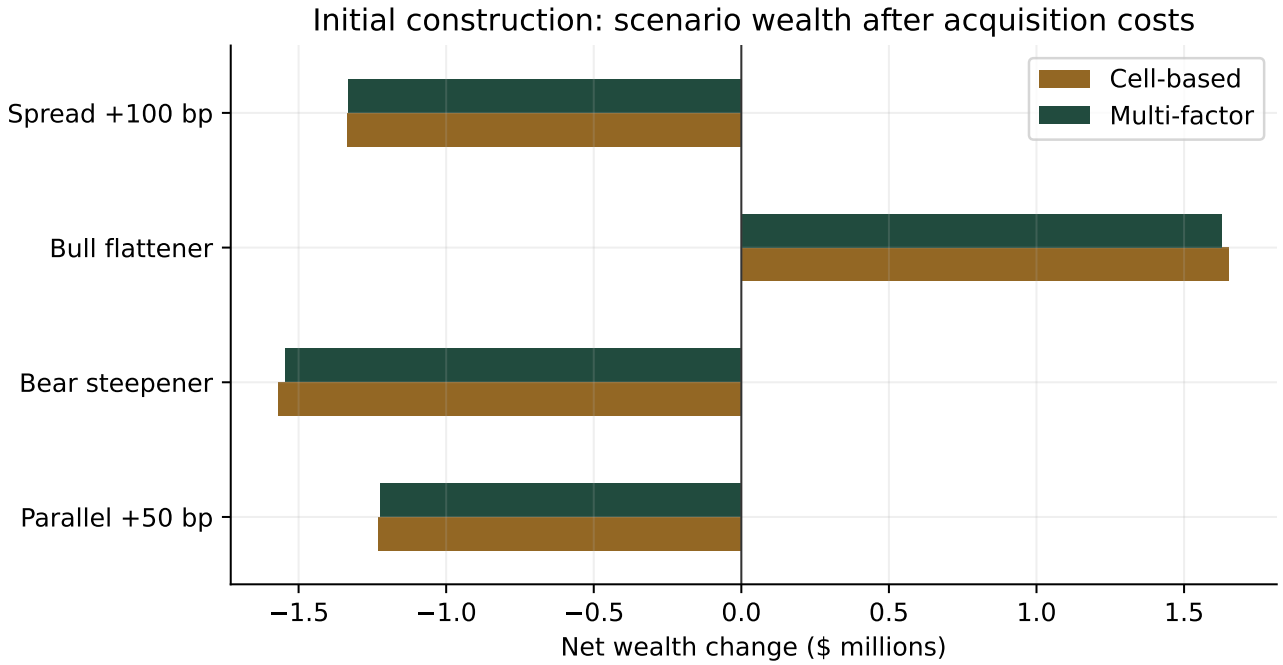
Gross price P&L applies to the actual amount invested in each bond sleeve. Net wealth change is terminal wealth minus initial cash: subtract the one-time purchase costs from that gross P&L. The benchmark is a costless reference index with the full $50 million invested. Neither table adds annual carry to an instantaneous shock.

Gross price P&L ($ millions)

Scenario	Benchmark	Cell-based	Multi-factor
Parallel +50 bp	-1.191	-1.202	-1.193
Bear steepener	-1.515	-1.539	-1.517
Bull flattener	1.657	1.683	1.657
Spread +100 bp	-1.284	-1.305	-1.302

Net wealth change after purchase costs ($ millions)

Scenario	Benchmark	Cell-based	Multi-factor
Parallel +50 bp	-1.191	-1.232	-1.223
Bear steepener	-1.515	-1.569	-1.546
Bull flattener	1.657	1.653	1.628
Spread +100 bp	-1.284	-1.335	-1.331



9 Explore construction choices and factor shocks

The HTML tools use the calculated allocations and risk matrix above. The PDF contains the corresponding tables and figures. These controls perform local calculations; they do not request market data or solve a new optimization.

9.1 Move between the two feasible construction plans

A weighted mixture of the two plans remains feasible because all mandate constraints are linear in bond-sleeve weights. The risk model evaluates each mixture; it need not be a linear average of the two endpoint tracking errors.

9.2 Change the four market factors explicitly

The default is a +50 bp parallel Treasury move and +25 bp common-spread widening. Set only one Treasury slider to a nonzero value to isolate a key-rate shock. Set all three Treasury sliders to the same value for a parallel shift.

10 Validate and interpret the decision

The calculations check that weights sum to one, no short positions appear, every common mandate limit passes, exact cells match under the cell method, the covariance matrices are positive definite, and acquisition budgets reconcile. The multi-factor portfolio is also checked to have no more modeled TE than the feasible cell portfolio under the same common mandate.

The cell method provides a transparent benchmark-bucket match and a low-cost selection rule. The multi-factor method explicitly accounts for co-movement and residual risk. Neither method is guaranteed to outperform. Before investing, verify security data, issuer relationships, executable prices, settlement costs, and feasible lot sizes, then rerun the constraints on the rounded purchase plan.

11 Mini project: raise the yield-proxy requirement

Raise the minimum weighted-yield increment from **2 bp to 5 bp above the benchmark**, keeping every other input and limit fixed.

1. Determine whether exact cell matching can meet the revised requirement.
2. Find the maximum attainable yield proxy under exact cells and the other limits.
3. Construct a multi-factor target under the revised floor, if feasible.
4. Report its annual TE, duration, sector weights, and purchase-budget reconciliation.
5. Explain which requirement would need to change if a method is infeasible.

Do not label an infeasible solution a portfolio. Do not interpret a higher yield proxy as a guaranteed expected excess return.

Worked solution: feasibility and the revised construction

The greatest exact-cell yield increment is **2.284 bp**, below the revised 5 bp floor. Exact cell matching is infeasible under the revised mandate.

Target	Yield proxy (%)	Duration (years)	Annual TE (bp)
Multi-factor, +5 bp floor	4.994	4.892	30.501

Target	Treasury (%)	Corporate (%)	Largest bond (%)
Multi-factor, +5 bp floor	25.00	50.00	18.00

Bond	Weight (%)	Initial buy $(m)\|Cost()$	
UST 2Y	0.000	0.000	0.000
UST 5Y	16.203	8.097	1619.338
UST 10Y	8.797	4.396	1318.744
Agency 3Y	18.000	8.995	2698.381
Agency 7Y	7.000	3.498	1748.951
Bank A 3Y	0.000	0.000	0.000
Bank B 5Y	0.000	0.000	0.000
Utility A 7Y	18.000	8.995	8095.143
Industrial A 10Y	1.828	0.913	1096.072
Industrial B 5Y	16.726	8.358	6686.522
Telecom A 10Y	0.000	0.000	0.000
Consumer A 7Y	13.446	6.719	6718.860

Bond values of **$49,970,017.99** plus costs of **$29,982.01** use exactly **$50,000,000.00** of initial cash.

The exact-cell approach would require a lower yield floor, different cell budgets, a broader investable universe, or another relaxed constraint. The multi-factor approach meets the revised floor by using its freedom across cells, with greater active risk than its original 2 bp target. That is a construction tradeoff, not evidence of a free improvement.